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TEXAS
The University of Texas at Austin

LONGHORN HAPKIT

Lab guide #3

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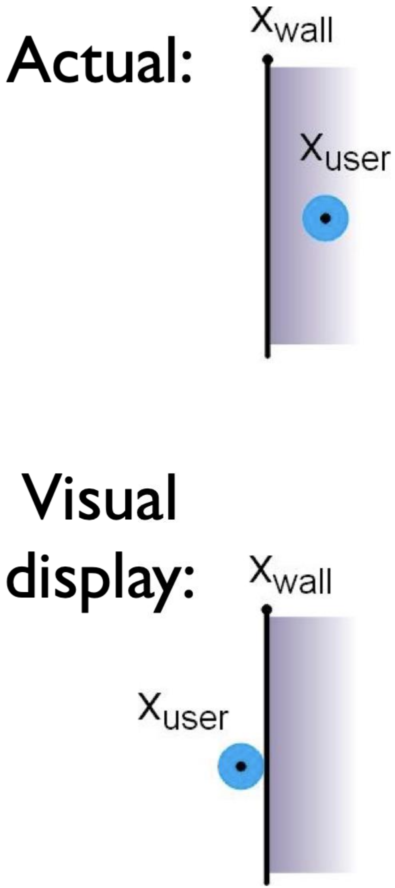
One more way to make a wall better: Visual Feedback of Stiffness

- **trick:** never show the point passing into the surface, even if it is!
- psychophysical studies have shown that this makes the surface appear stiffer to the user



Python!

- Let's use Python to interact with the hapkit
- Open a new terminal tab or window (the one running arduino needs to keep running arduino, don't mess with it)
- You should be in your home directory when you do this
  - The command "`pwd`" prints your working directory, which should read `/u/horns<yournumber>` if you are in your home directory
  - A shortcut back to your home directory if you ever need is "`cd ~`"

Virtual Environments

- A virtual environment is an isolated Python workspace for a specific project
- It contains its own Python interpreter and installed packages
- Packages installed in the venv do not affect the system Python installation
- You can make as many as you need
- If you want to share your project, you can create a dependency file (e.g., requirements.txt) so others can recreate the same environment
- On these computers you need to be in one to install new Python packages

Making a Virtual Environment

- In your terminal type

```
python3 -m venv <myvenv>
```

- This creates a new folder named <myvenv> with its own Python executable and its own directories for installed packages
- Activation scripts are installed
- In the terminal type

```
source <myvenv>/bin/activate
```

- You now have activated the virtual environment in this terminal
- You can download packages now

Download PySerial, Try Example

- With your activated environment install PySerial

```
pip install pyserial
```

- "pip" contacts the Python Package Index to find the pyserial package
- It then downloads the package files and any required dependencies and installs into the currently active Python environment

Try out an example

- On the github, download or copy/paste into a text editor the "hapkitCommands.py" file
- This file accompanies the spring_wall_damper_ISR.ino arduino sketch
- Run the arduino sketch
- In your terminal, navigate to the directory where you have placed "hapkitCommands.py" and type:

  ```
  python hapkitCommands.py
  ```
- You should now be able to command the hapkit from your terminal using python

Writing New Code

- You can write new python scripts in any text editor and save the file with a .py extension
- Looks like these computers have vscode, idle, vim, etc. so just take your pick

Resources

- Python tutorial https://docs.python.org/3/tutorial/index.html
  - Feel free to use whatever tutorial you find useful
- Navigating linux directories https://docs.ycrc.yale.edu/PIL/02-filedir/
- Python packages that may be useful
  - **tkinter** is great for building GUIs and basic games
  - **pygame** is more geared toward games, can do 3D
    - **pygame.mixer** can even add sound effects which can enhance haptic realism!
  - **matplotlib** for more scientific visualization, plotting
  - **pyqtgraph** good for live data streams
  - **numpy** for fast, advanced math operations
  - **scipy** for scientific computing (signal processing, filtering, optimization)
  - **pandas** for data logging – save files as CSVs and whatnot
- Some campers are also getting great results using HTML

Go forth and create

- This was a super simple example, but you can now use the vast world of python packages to create GUIs, games, link kits together, etc.
- Feel free to use LLMs, but please try and read the code they generate to understand what is going on
  - Vibe code your hearts out if you want *but*
  - Ask the LLM why things are the way they are!
- **Share** with the camp!
  - There's new directory in the github called camper_scripts
  - Upload your own folder with your required files and a readme to get people up to speed if they want to try it out on their own